UNIT 1: SOLUTIONS

Questions 1–25 · Integrated pilot

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QUESTION 1 [M] Define a solution and distinguish between a saturated, unsaturated and supersaturated solution.

ANSWER 1

A solution is a homogeneous mixture. Unsaturated solutions can dissolve more solute at the stated temperature; saturated solutions are at equilibrium with undissolved solute; supersaturated solutions contain more dissolved solute than the equilibrium solubility and are metastable.

QUESTION 2 [M] Explain why a solution of a non-volatile solute has a lower vapour pressure than the pure solvent.

ANSWER 2

For a non-volatile solute, only solvent contributes appreciably to vapour. Its mole fraction is below one, so Raoult's law gives $p_{solvent}=x_{solvent}\,p^\circ_{solvent}<p^\circ_{solvent}$.

QUESTION 3 [M] State Raoult's law for a solution containing two volatile components.

ANSWER 3

For ideal binary volatile liquids A and B: $p_A=x_{ApA}{}^\circ$, $p_B=x_{BpB}{}^\circ$ and $P_{total}=p_A+p_B=x_{ApA}{}^\circ+x_{BpB}{}^\circ$.

QUESTION 4 [H] Derive the relation between relative lowering of vapour pressure and mole fraction of a non-volatile solute.

ANSWER 4

For solvent A and non-volatile solute B, $p_A=x_{ApA}{}^\circ$. Thus $(p_A{}^\circ-p_A)/p_A{}^\circ=1-x_A=x_B$, the solute mole fraction. This assumes an ideal dilute solution and negligible solute vapour pressure.

QUESTION 5 [M] What is an ideal solution? Give two conditions for ideal behaviour.

ANSWER 5

An ideal solution obeys Raoult's law at every composition. Mixing has $\Delta H_{mix}=0$ and $\Delta V_{mix}=0$; unlike and like intermolecular interactions are approximately equal.

QUESTION 6 [H] Explain positive and negative deviations from Raoult's law with suitable examples.

ANSWER 6

Positive deviation occurs when A–B attractions are weaker than A–A/B–B, raising vapour pressure (e.g. ethanol–acetone approximately). Negative deviation occurs when A–B attractions are stronger, lowering vapour pressure (e.g. acetone–chloroform).

QUESTION 7 [M] Why do some liquid pairs show positive deviation from Raoult's law?

ANSWER 7

Unlike molecules attract one another less strongly than like molecules; they escape more readily to vapour, so the observed vapour pressure exceeds Raoult's-law prediction.

QUESTION 8 [H] A liquid mixture shows negative deviation from Raoult's law. Predict the signs of ΔHmix and ΔVmix and justify.

ANSWER 8

Negative deviation indicates stronger A–B attractions. Mixing is generally exothermic, $\Delta H_{mix}<0$, and contraction commonly gives $\Delta V_{mix}<0$.

QUESTION 9 [M] Define molarity, molality, mole fraction and mass percentage.

ANSWER 9

Molarity M=moles solute/L solution; molality m=moles solute/kg solvent; mole fraction $x_i=n_i/\Sigma n$; mass percent=100(mass solute/mass solution).

QUESTION 10 [H] Why is molality preferred over molarity when studying colligative properties?

ANSWER 10

Molality is based on solvent mass and is essentially independent of temperature, unlike molarity, whose volume basis changes with temperature.

QUESTION 11 [M] Calculate the molarity of a solution containing 5.85 g NaCl in 500 mL of solution.

ANSWER 11

n(NaCl)=5.85/58.44=0.1001 mol; V=0.500 L. M=0.1001/0.500=0.200 mol L^{-1}. Because "500 mL" has ambiguous trailing-zero precision, report 0.2, 0.20, or 0.200 M if it is intended to 1, 2, or 3 significant figures, respectively.

QUESTION 12 [H] A solution is prepared by dissolving 18 g glucose in 180 g water. Calculate the mole fractions of glucose and water.

ANSWER 12

n(glucose)=18/180=0.100 mol; n(H_2O)=180/18=10.0 mol. $x_{glucose}$=0.00990099… and x_{water}=0.990099…. If 180 g water is intended to 2 significant figures (matching 18 g glucose), report 0.0099 and 0.99; if 180 g is only 1 significant figure, round both to 1 significant figure.

QUESTION 13 [M] Define solubility and state the effect of temperature on the solubility of most solids in liquids.

ANSWER 13

Solubility is the amount of solute that dissolves in a specified amount of solvent at a stated temperature (and pressure for gases) to form a saturated solution. Most solids become more soluble as temperature rises, but exceptions occur.

QUESTION 14 [M] State Henry’s law and write its mathematical expression.

ANSWER 14

Henry’s law: at fixed temperature, dissolved-gas mole fraction is proportional to its partial pressure: $p=k_H\,x$.

QUESTION 15 [H] Explain why the solubility of a gas in a liquid generally decreases when temperature increases.

ANSWER 15

Gas dissolution is usually exothermic. Heating favours gas escape (Le Châtelier), so gas solubility generally falls as temperature rises.

QUESTION 16 [M] Why are soft drinks bottled under high pressure?

ANSWER 16

High CO_2 pressure increases its dissolved mole fraction by Henry’s law; opening the bottle lowers pressure and the gas escapes as bubbles.

QUESTION 17 [H] At a fixed temperature, explain how the mole fraction of a dissolved gas depends on its partial pressure.

ANSWER 17

At fixed temperature, Henry’s law gives $x=p/k_H$; dissolved-gas mole fraction rises linearly with partial pressure in the dilute range.

QUESTION 18 [M] What are solid solutions? Give one example of a substitutional solid solution.

ANSWER 18

A solid solution is a homogeneous solid mixture. Example: substitutional Cu–Ni alloy, where atoms of one metal replace atoms of the other in the lattice.

QUESTION 19 [M] Define colligative properties and list the four principal colligative properties.

ANSWER 19

Colligative properties depend on the number of dissolved particles, not their identity. The four are relative lowering of vapour pressure, boiling-point elevation, freezing-point depression and osmotic pressure.

QUESTION 20 [H] Why do colligative properties depend on the number of solute particles rather than their chemical identity?

ANSWER 20

In dilute ideal solutions, solvent chemical potential changes according to solute-particle mole fraction. Therefore effects scale with particle count; dissociation/association changes the count and is captured by i.

QUESTION 21 [M] Define relative lowering of vapour pressure.

ANSWER 21

Relative lowering is $(p^\circ-p)/p^\circ$, where p° is pure-solvent vapour pressure and p is solution vapour pressure. For an ideal solution with non-volatile solute it equals x_{solute}.

QUESTION 22 [M] Define elevation in boiling point and write its relation to molality.

ANSWER 22

Boiling-point elevation is $\Delta T_b = T_b(\text{solution}) - T_b^\circ$; for a dilute solution, $\Delta T_b = K_b\, m$ (or iK_{bm} for an electrolyte).

QUESTION 23 [M] Define depression in freezing point and write its relation to molality.

ANSWER 23

Freezing-point depression is $\Delta T_f = T_f^\circ - T_f(\text{solution})$; for a dilute solution, $\Delta T_f = K_f\, m$ (or iK_{fm} for an electrolyte).

QUESTION 24 [M] What is osmotic pressure?

ANSWER 24

Osmotic pressure π is the minimum external pressure applied to a solution to stop net solvent flow through a semipermeable membrane.

QUESTION 25 [H] Derive the equation π = CRT for a dilute solution and state the assumptions involved.

ANSWER 25

At equilibrium, $\mu_A^*(P) = \mu_A(\text{solution}, P+\pi) \approx \mu_A^*(P) + V_A\pi + RT \ln x_A$, so $\pi = -(RT/V_A)\ln x_A$. For a dilute solution, $-\ln x_A \approx x_B \approx n_B/n_A$ and $V \approx n_{AVA}$; hence $\pi V \approx n_{BRT}$, or $\pi = CRT$ with $C = n_B/V$. Assumptions: ideal dilute solution, small pressure correction, constant solvent partial molar volume, solvent-only membrane, uniform T, and negligible solute volatility.